

matchgroup_test_routed

Design Report

kicad-tools 0.13.0

Rev 1 | 2026-06-08 | jlcpb

Board Summary

Property	Value
Layers	4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu)
Footprints	8 (0 SMD, 0 THT, 8 other)
Nets	34
Traces	703 segments
Vias	41
Board Size	110.0 x 95.0 mm

Design Overview

Theory of Operation

Match-Group Test Board

N-trace + group-of-pairs match-group regression testbench

Epic #2661 Phase 3L (issue #2724)

Power Architecture

Power Rails: PWR_FLAG

Assembly Notes

5 fine-pitch components

- **Fine-pitch components:** 5 (U2, U1, U5, U4, U3)

ERC Status

Metric	Count
Errors	0
Warnings	0

Status: SKIPPED – ERC skipped by user request

Bill of Materials

Value	Package	Qty	References
ADDR_HDR	PinHeader_1x09_P2.54mm_Vertical	1	J3
FFC6_MIPi	FFC_6P_1.0mm	1	J1
HDMI19	HDMI_A_Receptacle	1	J2
BGA49_HDMI	BGA-49_5.0x5.0mm_Layout7x7_P0.5mm	1	U4
QFN24_MIPi	QFN-24-1EP_4x4mm_P0.5mm	1	U3
QFN48_DDR_CTRL	QFN-48-1EP_7x7mm_P0.5mm	1	U1
QFN48_DRAM	QFN-48-1EP_7x7mm_P0.5mm	1	U2
QFP48_SRAM	LQFP-48_7x7mm_P0.5mm	1	U5

DRC Status

Metric	Count
Errors	6
Warnings	12
Blocking	6

Status: FAIL ### Violations by Type

Violation Type	Count
connectivity	6
silkscreen_text_height	4
pad_grid	4
zone_unfilled	3
clearance_segment_via	2
diffpair_clearance_intra	2
clearance_pad_segment	1
clearance_pad_via	1
silkscreen_over_pad	1

Manufacturing Readiness

Verdict: NOT_READY

Action Items

- **[CRITICAL]** Fix 6 blocking DRC violations (clearance_segment_via (2), diff-pair_clearance_intra (2), clearance_pad_segment (1))
- **[OPTIONAL]** Verify zone fill in KiCad: 4 nets appear incomplete but may be connected via zone fills
- **[OPTIONAL]** Verify zone fill in KiCad for 3 zone-connected nets
- **[OPTIONAL]** Review 12 DRC warnings

Routing Status

Metric	Value
Signal Net Completion	87.1% (27/31)
Overall Completion	82.4%
Complete Nets	28 / 34
Zone-Connected Nets	3
Incomplete Nets	6
Unconnected Pads	162

Zone-Connected Nets

- +1V2
- +1V8
- GND

Unrouted Signal Nets

- DQ3
- MIPI_CLK_N
- MIPI_DAT0_N
- TMDS_D0_N

Unrouted Signal Nets

- DQ3
- MIPI_CLK_N
- MIPI_DAT0_N
- TMDS_D0_N

Cost Estimate

Metric	Per Board (estimated)
PCB Fabrication	~4.09 USD
Components (estimated)	~2.8 USD
Assembly (estimated)	~2.05 USD
Total (estimated)	~8.94 USD
Batch Quantity	5
Batch Total (estimated)	~44.72 USD